Network Traffic Classification & Attack Detection

Reviewer-updated project report

Current trained benchmark: BENIGN vs DDoS. The documentation now distinguishes this binary benchmark from the planned multi-attack CICIDS2017 extension.

1. Scope and dataset

The supplied raw dataset contains 225,745 rows and 79 columns. After exact duplicate-row removal, 223,112 rows remained. The bundled cleaned labels are DDoS (128,016) and BENIGN (95,096).

Important scope clarification: this raw file is the DDoS-specific CICIDS2017 capture slice. It therefore cannot train a model to classify attack types whose labels are absent from the file. True multi-attack classification requires combining the other CICIDS2017 attack files and retraining.

2. Modeling and evaluation

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	99.9933%	99.9961%	99.9922%	99.9941%	99.9935%
Random Forest	99.9933%	99.9961%	99.9922%	99.9941%	99.9961%
Extra Trees	99.9933%	99.9961%	99.9922%	99.9941%	99.9961%
XGBoost	99.9955%	100.0000%	99.9922%	99.9961%	99.9985%
Soft Voting Ensemble	99.9955%	100.0000%	99.9922%	99.9961%	99.9984%
Tuned XGBoost	99.9955%	100.0000%	99.9922%	99.9961%	99.9980%

The final evaluation is a group-aware internal holdout: identical cleaned numeric feature vectors are kept within one group. Feature selection is fit on the training portion only. This reduces one important leakage pathway but is not equivalent to external capture validation.

3. 99.99% accuracy - checked and qualified

The reported tuned XGBoost result is 99.995518% accuracy, with TN=19,019, FP=0, FN=2, TP=25,601. Recomputing accuracy directly from the confusion matrix gives 99.995518%, so the displayed figure is internally consistent.

The concern is therefore not a calculation error. The qualification is about generalization: the holdout still comes from the supplied capture family, and the source materials do not expose timestamp/session/flow-ID information required for a stronger temporal or session-based holdout. The number should be reported as a dataset-specific benchmark, not as guaranteed real-world performance.

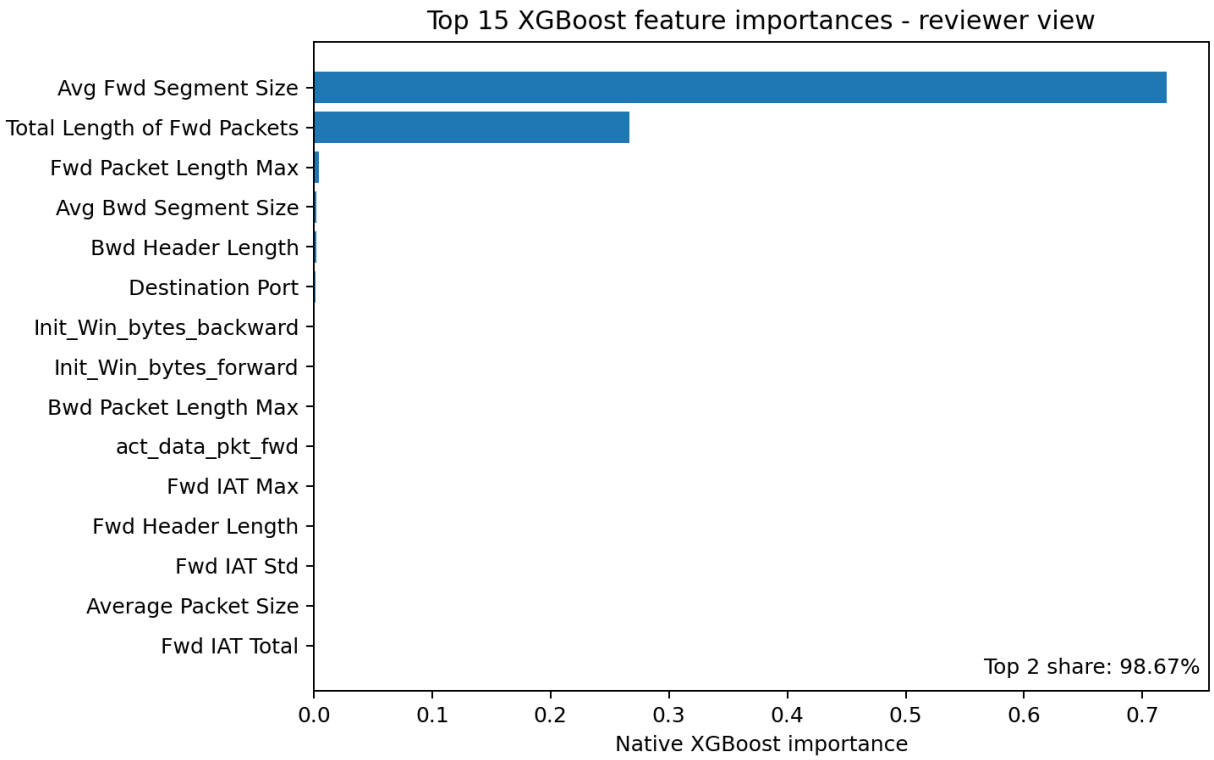
Bundled 6,000-flow sanity check (sample-level, not independent validation):

Feature set	5-fold accuracy mean	Std	5-fold F1 mean
Top 2 only	99.8500%	0.0972%	99.8500%
All 30 selected	99.9333%	0.0624%	99.9332%

This check confirms that the dataset remains highly separable even when the bundled sample is split into five folds, but it does not answer the external-generalization question.

4. Why do two features dominate?

The final deployment model was built from 30 selected features. Yet the native XGBoost importance chart assigns 98.67% of total importance to the top two features and only 1.33% to the other features combined.



A major reason is feature redundancy. In the bundled 6,000-flow sample, the following exact duplicate column pairs were observed:

- Total Length of Fwd Packets = Subflow Fwd Bytes
- Subflow Bwd Bytes = Total Length of Bwd Packets
- Avg Bwd Segment Size = Bwd Packet Length Mean
- Fwd Header Length = Fwd Header Length.1
- Avg Fwd Segment Size = Fwd Packet Length Mean
- Subflow Fwd Packets = Total Fwd Packets

This matters because tree-based native importance is not a unique measure of information content. When two variables encode the same signal, the model may split mostly on one and assign the other near-zero importance. The mutual-information ranking in the project shows that many other packet-length, timing, port, TCP-window and flow features still carry information about the label.

5. Reviewer-ready interpretation

The project does not claim that only two measurements matter for network security. It reports that two measurements dominate the current XGBoost gain on this particular binary dataset. For a revised training pass, exact duplicate columns should be removed before selection and feature contribution should be checked with permutation importance or SHAP in addition to native tree gain.

6. Multi-attack extension

The full CICIDS2017 release distributes different attack families across separate capture files. A multi-class version of this project would combine those files, keep the original attack labels, and train a multi-class XGBoost or ensemble model. Evaluation should emphasize macro-F1, per-class recall, class-wise support and a multi-class confusion matrix rather than relying on overall accuracy alone.

Target output of the extension: BENIGN, DDoS, PortScan, Bot, Infiltration, Web Attack variants, FTP-Patator, SSH-Patator and the relevant DoS variants present in the combined data. The current bundled binary model should not be presented as already performing this multi-class task.

7. Final limitations and next step

The dashboard, API and deployment packaging remain usable for the current binary benchmark. The next evidence milestone is independent validation on a separate contemporary network capture, followed by multi-attack retraining and class-wise evaluation. This will turn the current high-scoring benchmark into a more defensible claim about generalization.

Source note: CICIDS2017 is distributed as multiple capture-day CSVs with different attack families; the current project uses the DDoS-specific slice. The accompanying project ZIP does not include the other raw attack files, so they were not silently fabricated or substituted during this update.